LITERATURE SURVEY

Agentic Incident Response and Trustworthy Evidence Fusion

Project Title	Trust-Aware Agentic AI for Autonomous Cyber Incident Response and Resilient Recovery
Student	Dhruva Hegde
Team	Team 221
Institution	PES University, Bengaluru
Survey Scope	Two research papers published in 2026

Prepared as an individual literature survey for the research component of the Team 221 capstone project.

1. Introduction

The capstone project, **Trust-Aware Agentic AI for Autonomous Cyber Incident Response and Resilient Recovery**, investigates how an AI-assisted system can support cyber incident handling while accounting for operational risk and the trustworthiness of recovery evidence. The project connects two research areas: adaptive incident response that evaluates the consequences of defensive actions, and quantitative trustworthiness assessment that combines evidence from multiple sources, including evidence relevant to recoverability.

This survey reviews two complementary 2026 studies. The first examines an agentic incident-response framework that validates response actions before execution and re-evaluates plans as an attacker adapts. The second proposes a multi-source evidence-fusion method for software trustworthiness assessment.

Scope note: Neither paper defines Team 221's exact Recovery Trust Score (RTS). The second paper offers a transferable evidence-fusion foundation; adapting it to backup integrity, recovery-point freshness, malware-free validation, provenance, and restore-test outcomes would be a project-specific contribution.

2. Research Paper 1: Adaptive Agentic Incident Response

Title: An adaptive agentic AI framework for counterfactual validation and adversarial lookahead in automated incident response

Authors: Sheetal Prasad Kizhakel and Sagar Dhanraj Pande
Publication: *Discover Artificial Intelligence*, vol. 6, Article 970, 2026
DOI: 10.1007/s44163-026-02058-6 | Springer Nature, Open Access

Problem addressed

Traditional Security Orchestration, Automation and Response (SOAR) systems commonly depend on predefined playbooks. LLM-assisted systems generate more flexible recommendations, but a plan generated when an alert first arrives may be executed without predicting the side effects of individual actions. Further, attackers can adapt after defensive actions, making later steps in the original plan unsafe or ineffective.

Proposed methodology

The authors introduce **CIPHER-A** (Counterfactual Incident Response Planning with Hallucination-Aware Evaluation and Reasoning – Adaptive), a closed-loop agentic framework combining response planning, pre-execution consequence assessment, attacker-state prediction, and repeated validation.

- **Response-plan generation:** an LLM proposes incident-response actions using the observed alert and environment context.
- **Counterfactual Validation Engine (CVE):** evaluates possible outcomes of a proposed action before execution. It generates distinct outcome scenarios and computes a validation score considering action risk, asset criticality, and response urgency.
- **Decision routing:** depending on the validation result, an action may proceed, be modified to a safer alternative, or be escalated to a human analyst.
- **Attacker Simulation Agent (ASA):** predicts how the adversary may change tactics after an executed action, grounded in MITRE ATT&CK.;
- **Sequential re-validation:** remaining actions are checked again against the updated threat state instead of blindly continuing with the original plan.

Technologies, algorithms, and datasets

The paper reports Claude API and Mistral-7B as language-model engines, LangGraph for agent orchestration, Wazuh for SIEM alert ingestion, CrowdStrike, AWS and Active Directory APIs for response execution, and PostgreSQL for audit logging. MITRE ATT&CK; grounds attacker-behaviour reasoning. Evaluation uses DARPA Operationally Transparent Computing (OPTC), MITRE ATT&CK; scenarios, and the CTU-13 botnet dataset.

Key findings and results

The authors report evaluation across 143 incidents and three public dataset/scenario sources. Compared with baseline SOAR, the framework reports a **59.2% relative reduction in Response Plan Degradation Rate (RPDR)**, a **3.2% false-safety rate** versus a reported 22.7% baseline, **91.7% escalation accuracy**, and a **37.8% reduction in mean time to contain**. These results support validating actions before execution and revisiting the plan as the attacker state changes.

Limitations and research gaps

- The authors identify future work around knowledge-graph representations of network topology and asset dependencies to improve blast-radius analysis.
- Counterfactual outcome estimates rely on LLM-based probabilistic reasoning. The paper proposes integrating formal causal models and stronger verification to produce more definitive consequence predictions.
- Reported performance should not automatically be extrapolated to very large or highly concurrent enterprise environments. The paper notes the need for load testing of throughput, tail latency, queue growth, and accuracy when topology context is reduced.
- The paper focuses on incident-response action safety and attacker adaptation. It does not provide a dedicated recovery-point trust score, backup-integrity assessment, or dependency-aware restoration workflow.

Relevance to Team 221

This study is a strong architectural reference for the agentic response portion of the capstone. Team 221 can adapt its closed-loop principle so proposed containment and recovery actions are assessed before execution, and subsequent actions are reconsidered when new evidence changes the incident state. Its human-escalation mechanism supports policy gates for high-impact actions. The project can extend beyond the paper by applying similar validation to recovery operations, including selecting a restore point and restoring services in dependency order.

3. Research Paper 2: Multi-Source Trustworthiness Evidence Fusion

Title: A trustworthiness assessment for software based on multi-source evidence fusion

Authors: Weina Zhang, Dalin Zhang, Jintao Liu, and Lianying He
Publication: *Reliability Engineering and System Safety*, vol. 266, Article 111842, 2026
DOI: 10.1016/j.ress.2025.111842 | Elsevier, Open Access

Problem addressed

Trustworthiness assessment combines measurements from different sources and lifecycle stages. Direct operational evidence and indirect process/design evidence may have different meanings, quality levels, and uncertainty. Treating all evidence as homogeneous can conceal contradictions, overvalue weak sources, and produce unreliable estimates. The paper addresses classification, conflict resolution, and fusion of heterogeneous evidence.

Proposed methodology

The authors propose a multi-stage evidence-fusion framework that distinguishes two evidence categories and resolves conflicts before combining them into an overall assessment.

- **Evidence classification:** direct evidence is derived from actual operation and outcomes; indirect evidence comes from development/process artefacts and serves as a proxy for system properties.
- **Utility transformation:** measurements with different scales are mapped to a common utility scale and represented as Basic Belief Assignments (BBAs), enabling comparison and combination.
- **Heterogeneous evidence conflict:** falsity is used to identify conflicts between direct and indirect evidence, whose semantics differ.
- **Homogeneous evidence conflict:** credibility and uncertainty are used to address inconsistencies among evidence items of the same type.
- **Weight optimisation and fusion:** a game-theory-based linear combination determines weighting/discounting factors; adjusted evidence is combined using Dempster's rule from Dempster–Shafer evidence theory.

Technologies, algorithms, and data

The paper is a mathematical assessment and evidence-fusion method rather than a product built around a particular programming stack. Its core techniques are utility theory, Dempster–Shafer evidence theory, Basic Belief Assignments, falsity-based conflict measurement, credibility and uncertainty metrics, game-theoretic weight combination, and Dempster's rule. The experimental study uses three complementary sources from a real software system: lifecycle traceability documentation, standardized laboratory test data, and environmental simulation data. Demonstrated attributes include reliability, security, functionality, efficiency, and maintainability where sufficient evidence is available.

Key findings and results

The experiments report that the proposed approach satisfies monotonicity, acceleration, sensitivity, and substitutability. The authors also report improved accuracy and robustness over compared state-of-the-art evidence-fusion algorithms, particularly under evidence conflict, uncertainty, and bias. The reliability case study includes a recoverability sub-attribute, with indicators such as backup data rate, recovery originality, recovery-time satisfaction, recovery requirement design, and recovery matching consistency. This makes the study relevant to an RTS-like assessment, although its model is not itself a ransomware recovery-point score.

Limitations and research gaps

- The assessment is demonstrated using a real software-system case study and selected attributes; evidence coverage is not complete for every attribute. The paper's assessability table marks some attributes as not assessable because adequate evidence was unavailable.
- The model depends on the selection, quality, and calibration of input indicators, evidence credibility, uncertainty estimates, and expert-derived ground truth. Poor or manipulated input evidence can still affect the final assessment.

- The paper does not define cyber-specific indicators for backup immutability, cryptographic integrity, malware contamination, attacker dwell time, snapshot provenance, or restore-point isolation.
- The fusion method does not itself orchestrate incident response, determine service restoration order, or validate that a restored service is operational. These functions require integration with a separate response and recovery workflow.

Relevance to Team 221

This paper provides a strong conceptual foundation for the project's Recovery Trust Score (RTS). Team 221 can classify recovery evidence into direct observations (integrity-check results, restore-test outcomes, malware-scan results, and observed recovery time) and indirect evidence (backup policy, snapshot provenance records, configuration history, and dependency metadata). Evidence can be normalised, weighted according to credibility and uncertainty, and fused into a trust estimate. The project-specific extension is to define cyber-recovery indicators and map the score to a transparent decision: approve a recovery point, quarantine it for further validation, or reject it.

4. Comparative Analysis

Dimension	Paper 1: CIPHER-A	Paper 2: Evidence fusion
Primary focus	Adaptive agentic incident response and safe action execution	Trustworthiness assessment from heterogeneous evidence
Main problem	Unsafe/static response plans as attackers adapt	Conflicting evidence from different sources and lifecycle stages
Core mechanisms	LLM planning, counterfactual validation, attacker simulation, sequential re-validation	Direct/indirect evidence, utility mapping, credibility, uncertainty, game-theoretic weighting, Dempster's rule
Human oversight	Escalation when action validation indicates risk	Expert-derived ground truth and evidence-quality inputs
Recovery relevance	Can validate recovery actions, but no dedicated recovery-point score	Includes recoverability indicators, but no ransomware-specific restore-point workflow
Contribution to Team 221	Agentic response, policy gates, repeated action validation	RTS evidence classification, weighting, and fusion

The papers are complementary rather than interchangeable. CIPHER-A contributes a control loop for deciding whether and when an action should be executed. The evidence-fusion paper contributes a method for estimating how much confidence should be placed in diverse evidence. A combined design can use evidence assessment to inform action validation and escalation decisions.

5. Identified Research Gap and Proposed Project Direction

The reviewed studies address important parts of the problem, but neither provides an end-to-end framework connecting incident understanding, trustworthy recovery-point selection, dependency-aware restoration, and post-restoration validation. CIPHER-A focuses on validating defensive actions and modelling attacker adaptation. The evidence-fusion study focuses on trustworthiness assessment across heterogeneous evidence, with recoverability as one software attribute. Integrating these ideas for cyber incident recovery remains a project opportunity.

Proposed project-specific direction

- **Recovery evidence model:** define measurable indicators for backup age/freshness, cryptographic integrity, immutability, provenance, malware-scan status, isolation, restore-test success, and recovery-point consistency.
- **Evidence trust and fusion:** distinguish observed/direct evidence from indirect metadata and policy evidence; account for source credibility, uncertainty, and conflicts before calculating RTS.
- **Decision thresholds:** translate RTS into transparent decisions such as trusted, needs additional validation, or rejected. Calibrate thresholds through experiments rather than selecting them arbitrarily.
- **Agentic response control:** use an action-validation layer before high-impact operations and re-evaluate the plan when new incident or recovery evidence arrives.
- **Dependency-aware recovery:** represent service and asset dependencies so critical services can be restored in a safe and logical order.
- **Recovery verification:** validate restored service health, security status, and consistency before marking recovery complete.

Research gap statement: Existing work separately advances adaptive agentic incident response and multi-source trustworthiness assessment, but does not fully integrate cyber-specific recovery-point trust evaluation with policy-controlled, dependency-aware restoration and post-recovery validation in one workflow.

6. Conclusion

The two selected 2026 studies provide a relevant foundation for Team 221's capstone. Kizhakel and Pande demonstrate how an agentic response system can validate the consequences of actions and adapt to attacker behaviour during execution. Zhang and colleagues demonstrate how heterogeneous evidence can be classified, conflict-adjusted, weighted, and fused to produce a more robust trustworthiness assessment. The first study informs the project's agentic response and safety-control architecture; the second informs the design of the Recovery Trust Score. The capstone can build on both by developing and evaluating a cyber-recovery-specific trust model and integrating it with controlled, dependency-aware recovery operations.

7. References

[1] S. P. Kizhakel and S. D. Pande, “An adaptive agentic AI framework for counterfactual validation and adversarial lookahead in automated incident response,” *Discover Artificial Intelligence*, vol. 6, article 970, 2026. doi: 10.1007/s44163-026-02058-6. https://doi.org/10.1007/s44163-026-02058-6

[2] W. Zhang, D. Zhang, J. Liu, and L. He, “A trustworthiness assessment for software based on multi-source evidence fusion,” *Reliability Engineering and System Safety*, vol. 266, article 111842, 2026. doi: 10.1016/j.ress.2025.111842. https://doi.org/10.1016/j.ress.2025.111842

Source note: This survey was prepared from the two full-text research PDFs supplied for review. Reported performance figures are reproduced as stated by the papers and are not independent replications. Project-specific gaps are analytical observations based on comparison with Team 221's stated scope.